DABIGATRAN

Swallow capsules whole

Reversal: idarucizumab

CHEMICAL NAME Ethyl N-[(2-{[(4-{N′-[(hexyloxy)carbonyl]carbamimidoyl}phenyl)amino]methyl}-1-methyl-1H-benzimidazol-5-yl)carbonyl]-N-(pyridin-2-yl)-β-alaninate (dabigatran etexilate mesilate)

CLASS Direct oral anticoagulant (DOAC) — direct thrombin inhibitor

TRADE NAMES Pradaxa · Pradax · Prazaxa

THERAPEUTIC USES / INDICATIONS

- Stroke prevention in non-valvular atrial fibrillation.
- Treatment of DVT/PE (after 5 days of parenteral anticoagulant) and prevention of recurrence.
- VTE prophylaxis after hip/knee replacement.

ADMINISTRATION

- **PO** capsules swallowed whole with water — never open, crush or chew (↑ absorption 75% → bleeding).
- Take with food if dyspepsia; keep capsules in original blister (moisture-sensitive).

DOSAGES

- **AF:** 150 mg twice daily; 110 mg twice daily if ≥80 yr, on verapamil, or high bleeding risk (75–80 yr, CrCl 30–50, gastritis).
- **DVT/PE:** 150 mg twice daily after ≥5 days of LMWH.
- **Hip/knee:** 110 mg 1–4 h post-op, then 220 mg once daily.
- Contraindicated if CrCl <30.

PHARMACOKINETICS

- **Onset:** 0.5–2 h (peak). **Duration:** 12–24 h.
- Prodrug; bioavailability ~6.5%; P-gp substrate; not CYP-metabolised.
- 80% excreted unchanged in urine — accumulates in renal impairment. **t½:** 12–17 h.

PHARMACODYNAMICS

- Directly and reversibly inhibits **free and clot-bound thrombin (factor IIa)**.
- Prolongs aPTT and thrombin time.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspepsia, abdominal pain (tartaric acid core).
- Bleeding (minor).
- Nausea, diarrhoea.

ADVERSE EFFECTS (SERIOUS)

- Major bleeding (higher GI bleeding than warfarin at 150 mg).
- Spinal/epidural haematoma.
- Reversal: idarucizumab (Praxbind); dialysable.

CONTRAINDICATIONS & CAUTIONS

- CrCl <30 mL/min.
- Active significant bleeding; mechanical heart valves.
- Use with strong P-gp inhibitors (ketoconazole, ciclosporin, itraconazole, dronedarone).
- Hepatic impairment expected to affect survival; pregnancy.

KEY INTERACTIONS

- **P-gp inhibitors** (verapamil, amiodarone, ticagrelor, clarithromycin): ↑ levels — dose adjust/avoid.
- **P-gp inducers** (rifampicin, carbamazepine, St John's wort): avoid.
- **Antiplatelets, NSAIDs, SSRIs:** ↑ bleeding.

NURSING MANAGEMENT

- Check CrCl at baseline and at least yearly (more often if >75 yr or impaired).
- Give capsules whole; do not use via NG tube.
- Monitor for bleeding and dyspepsia.
- Peri-procedure stop times depend on renal function.

PATIENT EDUCATION

- Swallow capsules whole with water; keep them in the original pack.
- Indigestion is common — take with food; report if severe.
- Report bleeding; carry alert card.
- Do not stop without advice.

KEY POINTS

- Direct thrombin inhibitor.
- Never open capsules.
- Mostly renal; avoid if CrCl <30.
- Reversal: idarucizumab.